Build a complete prediction

antocciphysics | TEACHER KEY | CPA

Learning target: Connect net force, velocity change, and final position in one model.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 17; 3 course days later.*
At rest: friction 20 N opposite the push. Sliding: net force 20 N with the push.

02. Explain a worked example (worked)
Triangle base 2 s, height 4 m/s, area 4 m; net force arrow right.

03. Start with forces (model)
Net 4 N right; a 2 m/s² right.

04. Predict motion (practice)
vf 6 m/s right; average 3 m/s right;Δx 9 m right; xf 10 m.

05. Explain a limit (reason)
Changing push/friction, etc. More opposing friction reduces net force and acceleration for fixed mass.

06. Choose a method (mixed) *Review: Unit 1, Day 8; 12 course days later.*
a=2.8 m/s² east; average velocity 7 m/s east; displacement 35 m east.

07. Exit check (exit)
a 3 m/s² right; vf 6 m/s right; constant force/mass. Assessment looks at justified method and assumption.

08. STRETCH: Apply it another way (stretch)
Same acceleration 2 m/s², mass 4 kg, net force 8 N.

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01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 17; 3 course days later.*
N=29.4 N; static limit 14.7 N. Low push: friction 6 N, a=0. High push: kinetic friction 11.76 N, a=2.08 m/s².

02. Explain a worked example (worked)
Triangle base 2 s, height 4 m/s, area 4 m; net force arrow right.

03. End point versus displacement (model)
Endpoint (-20, 60) km; displacement (80, 60) km, magnitude 100 km; separation√4000=63.2 km.

04. Add bounds (practice)
δx=1+20(.1)=3 km→[-23,-17];δy=2 km→[58, 62].

05. A new force (reason)
a=.2 m/s²; extraΔv=2 m/s; extraΔx=10 m.

06. Choose a method (mixed) *Review: Unit 1, Day 8; 12 course days later.*
Stops after 7 s at distance 122.5 m. Extending to 9 s predicts -10 m/s, which assumes continued acceleration after stopping and reversal.

07. Exit check (exit)
Should report (-20, 60) km with x[-23,-17], y[58, 62]; limitations gravity/Earth motion/3 D, etc. A bound is not a probability.

08. STRETCH: Apply it another way (stretch)
a=3 m/s², v_f=10 m/s, x_f=3-8+24=19 m. Turns at 2/3 s; cannot use speed as a signed velocity.